

Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy

Abstract

Federated learning (FL) has become a central paradigm for privacy-aware machine learning because it permits collaborative model training without directly centralizing raw data. Yet the last several years have made clear that keeping data local is not, by itself, a formal privacy guarantee: gradients, parameter deltas, logits, and intermediate representations can still leak membership, labels, attributes, and even high-fidelity reconstructions of sensitive inputs. Differential privacy (DP) has therefore emerged as the dominant formal language for bounding information leakage in FL, while secure aggregation and related cryptographic mechanisms have become the dominant systems tools for reducing the trust placed in the server or the network. The resulting research area sits at the intersection of optimization, statistics, distributed systems, cryptography, and applied machine learning [1]-[30].

This survey reviews privacy-preserving machine learning through the combined lens of federated learning and differential privacy. It synthesizes the literature around privacy granularity, trust assumptions, privacy accounting, optimization design, personalization, communication efficiency, hybrid cryptographic protection, and application-specific adaptations for vertical FL, split learning, graph learning, recommender systems, healthcare, and large language models. A unifying theme is that the right question is not whether “FL is private,” but rather **what is protected, against whom, under which trust assumptions, and at what utility cost**. The survey argues that the field is moving from first-generation “add Gaussian noise to FedAvg” designs toward richer systems that co-design privacy accounting, adaptive clipping, secure aggregation, compression, personalization, and parameter-efficient fine-tuning. At the same time, the literature still faces unresolved challenges in tight and interpretable privacy accounting under realistic participation, auditing and empirical privacy validation, robustness under malicious behavior, and deployment for foundation models and structured cross-silo workloads [31]-[198]. ¹

Introduction

Federated learning was introduced to exploit decentralized data while avoiding direct raw-data pooling, and it quickly became influential in mobile and cross-silo environments where data ownership, bandwidth, regulation, or institutional rivalry prevent centralization [1]-[4]. Early FL work, however, was often interpreted too optimistically: although data remain on device or within organizational silos, the communicated updates are themselves rich statistical objects. Subsequent work demonstrated that collaborative and federated training pipelines can leak sensitive information through membership inference, property inference, label leakage, gradient inversion, and model inversion attacks [7]-[14]. This realization shifted the field from “data locality” toward formal privacy guarantees and explicit threat modeling. ²

Differential privacy offers a mathematically rigorous notion of indistinguishability that can be applied at different protection units: the record, the user, the client, or the silo. In FL, these distinctions are crucial. Record-level DP can protect individual examples within a client’s local dataset, but may not hide whether an entire client participated. Client-level or user-level DP instead seeks to hide the contribution

of a whole client or user across rounds, which is the more natural goal in many cross-device systems. Local DP removes trust in the server but usually sacrifices more utility; shuffle and distributed models recover some utility by exploiting anonymity or secure aggregation; trust-minimized cross-silo variants lie between central and local extremes [15]-[30], [80]-[102]. Terminological inconsistencies still persist across the literature, but the recent trend is toward more precise formulations based on neighboring relations and on what the adversary can observe. ³

The literature has now matured enough that multiple surveys have attempted to organize it, including surveys on differentially private federated learning, privacy-preserving FL strategies, FL security, and DP in broader machine learning [5], [6], [195]-[198]. What remains useful, however, is a synthesis that treats FL and DP not as isolated modules but as a **joint design problem** involving optimization, privacy accounting, trust reduction, communication, and model adaptation. This matters because the performance of private FL depends not only on the noise multiplier, but also on clipping geometry, participation schedules, local epochs, personalization structure, model architecture, and whether secure aggregation or compression changes the effective observable interface [29], [37]-[45], [103]-[120]. ⁴

The present survey therefore adopts four organizing principles. First, it distinguishes privacy notions by **where randomization occurs** and **what contribution is hidden**. Second, it groups methods by algorithmic family rather than by venue or application alone. Third, it emphasizes the role of **hybrid systems**, where DP, secure aggregation, homomorphic encryption, or trusted execution environments serve complementary rather than interchangeable roles. Fourth, it treats evaluation methodology, empirical attacks, and deployment constraints as central scientific questions rather than afterthoughts. This perspective is especially important in light of production-scale deployments of user-level private FL, including Gboard language models, and in light of rapidly growing work on private parameter-efficient fine-tuning for large and multimodal models [44], [73], [167]-[175]. ⁵

Methods

Foundations, threat models, and privacy granularity. The methodological core of private FL begins with defining the unit of protection and the adversary’s observational power. Standard central DP assumes a trusted curator (or at least a trusted mechanism applied to sufficient statistics); local DP assumes each client perturbs its own output before transmission; shuffle privacy assumes a randomized local report followed by anonymity or anonymization; federated f-DP and related formulations refine Gaussian-style accounting for the repeated, client-participatory structure of FL [15]-[30]. In FL specifically, one must further specify whether neighboring datasets differ by a record within a client, by an entire client, or by a user aggregated through a device. These choices induce different clipping operators, different sensitivity analyses, and different privacy-accounting regimes. Client-level DP typically clips each client update before adding calibrated noise to the aggregate; record-level DP often uses local DP-SGD-style clipping within client optimization; inter-silo and trust-minimized variants additionally restrict who may observe intermediate updates [31]-[45], [80]-[102]. ³

Central and user-level private federated optimization. The dominant algorithmic line in private FL adapts the DP-SGD template to decentralized training: clip an update, aggregate, add noise, and account for cumulative privacy loss. Foundational works framed this at client level [31], [32], extended it with performance analysis [33], [34], and then tightened optimization and accounting through heterogeneous-data analyses [37], [38], adaptive clipping [39], adaptive or structured noise [55]-[57], refined user-contribution control [52], improved accounting [17]-[21], and alternatives to sampling-based amplification such as DP-FTRL and streaming/matrix-mechanism perspectives [40]-[45]. A major practical milestone was the deployment of differentially private Gboard language models, which showed that user-level DP in FL can reach production utility when combined with public pretraining, adaptive

clipping, and careful participation control [44]. Empirical auditing and privacy evaluation have become a parallel line of work, demonstrating both the value and the limitations of nominal (ϵ, δ) -DP claims when implementations, clipping heuristics, or attack surfaces deviate from theoretical assumptions [45]-[48].

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A second thread within central/user-level methods focuses on **recovering utility**. This includes adaptive clipping [39], [70], personalized or transformed updates [64]-[69], regularization and sparsification [54], compression-aware private learning [62], instance-aware contribution smoothing [52], scale-sensitive and activation-design methods inherited from non-federated DP deep learning [49], [50], and selective update or release strategies [173]. Distillation-based families, including PATE-inspired and federated knowledge transfer methods, aim to move privacy costs away from end-to-end noisy gradient descent toward less sensitive teacher aggregation or transfer interfaces [75]-[79], [185]. These approaches are especially valuable when dimensionality is high, participation is sparse, or the model contains large embedding layers. The recent move toward parameter-efficient fine-tuning extends the same logic: privatize a small, structured set of trainable parameters rather than the full model [72], [73], [168]-[171].

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Local, shuffle, and trust-minimized private FL. When the server is not trusted to observe even uncloaked client updates, local perturbation becomes attractive. LDP-Fed and its successors privatize client outputs before transmission and often combine sign-based selection, top- k , personalization, clustering, adaptive budget allocation, or staircase/randomized-response mechanisms to partly recover utility [80]-[93]. The price is statistical: local privacy usually requires far more noise than central privacy for the same protection goal. This motivated shuffle-model approaches for FL, where client randomization is combined with anonymity or randomized routing so that privacy amplification compensates for weaker local guarantees [94]-[99]. Related trust-minimized work studies FL without a trusted server, inter-silo record-level privacy, private non-convex learning without a trusted server, and decentralized random-walk learning where privacy depends on communication topology rather than a single aggregator [100]-[102]. Taken together, these papers show that “central DP versus local DP” is too coarse a dichotomy; the actual design space spans a continuum of trust assumptions and amplification mechanisms. 8

Cryptographic hybrids and systems co-design. Differential privacy and secure aggregation solve different problems. DP bounds inferential leakage from outputs; secure aggregation hides individual messages in transit and at the server interface. Consequently, the strongest recent systems treat them as complementary. Practical secure aggregation in FL began with pairwise-mask protocols [103], while related privacy-preserving ML systems used secure multiparty computation [104] or crypto-assisted DP [105]. Later work quantified the communication-privacy trade-off induced by secure aggregation [109], developed discrete-noise mechanisms friendly to modular arithmetic such as distributed discrete Gaussian, Poisson-binomial, and Skellam mechanisms [110]-[112], and combined private learning with autotuned or distribution-aware compression [118]-[120]. Fully or selectively homomorphic variants such as BatchCrypt and FedML-HE demonstrate that cryptography can be engineered into cross-silo FL, although overhead and model-size growth remain significant [106], [107]. Newer work investigates verifiability, robustness, and scalable multi-server topologies—for example, DDP-SA and verifiable-DP systems—suggesting that future FL stacks may integrate DP, secure aggregation, compression, and correctness proofs in a single pipeline [113]-[117]. 9

Structured, vertical, transfer, and split settings. Most early DP-FL papers studied horizontal FL, where clients share a feature space and differ in samples. But real deployments frequently involve vertically partitioned data, distributed features, federated transfer, or split computation. FDML and VAFL were early systems for distributed-feature collaboration [121], [122], while secure federated transfer learning and

HELEN addressed cross-party learning under limited overlap [123], [124]. DP-specific work on VFL and tree-boosting includes hybrid private training on vertically partitioned data, privacy-preserving tree ensembles, FederBoost, OpBoost, and robust/privacy-aware split-VFL pipelines such as FedVS [125]-[130]. This branch is particularly important because privacy leakage often shifts from gradient inversion to **label leakage, feature leakage, or entity-resolution leakage**. Indeed, recent attacks and defenses in VFL and split learning show that interface design—not just parameter noise—determines what can be inferred [131]-[140]. The implication is methodological: DP in structured FL must often be combined with architecture-aware masking, party-specific representation control, or secure entity matching rather than naively ported from horizontal FedAvg. ¹⁰

Applications and model families. The field has grown from image-classification benchmarks toward application-specific private FL pipelines. Healthcare and biomedicine remain the flagship domain because they combine strong privacy requirements, distributed data ownership, and clear incentives for collaboration. Representative lines include federated EHR prediction, medical-imaging FL, sensitive-health-data DP-FL, private medical-data training, and privacy-preserving omics [141]-[146]. Recommendation and personalization form a second major branch, spanning collaborative filtering, matrix factorization under DP or LDP, federated recommendation with graph models, and cross-silo recommendation distillation [147]-[152]. Finance, vehicular and urban sensing, energy, Bayesian optimization, LDA, and graph analytics provide additional examples where sensitive local data coexist with strong dependence on local heterogeneity [153]-[165]. These application papers repeatedly show that privacy-utility trade-offs are not architecture-invariant: tabular, tree, sequence, graph, and recommendation workloads respond differently to clipping, local steps, and the privatization locus.

The newest application frontier is foundation models. For large language or multimodal models, direct end-to-end private FL is often impractical because embedding layers and adapters are high-dimensional and sensitive to clipping. Recent work therefore emphasizes private parameter-efficient tuning, private LoRA, federated prompt learning, public-model assistance, and selective privacy mechanisms for LLM adaptation [167]-[175]. DP-FPL extends this to multimodal prompt tuning with mixed global and local privacy treatment [167], while DP-LoRA and FedASK show how low-rank structure can reduce communication and sometimes improve privacy-utility behavior [168], [169]. Production-facing reports from Gboard suggest that public pretraining and strong systems engineering are indispensable for high-utility private language modeling [44], [175]. More broadly, foundation-model work makes clear that private FL is increasingly a problem of **what to fine-tune** rather than simply **how much noise to add**. ¹¹

Optimization, personalization, and evaluation infrastructure. Private FL cannot be separated from the broader optimization and systems context of FL. Heterogeneous-data baselines such as SCAFFOLD, FedBuff, FedPAQ, FedProx, meta-learning and Moreau-envelope personalization, client selection, and asynchronous federated optimization all influence how clipping and noise propagate through the training trajectory [176]-[191]. Production examples such as keyboard prediction and Smart Compose show that private FL is not a laboratory curiosity but a deployed systems problem [192], [193]. At the same time, benchmarking and reproducibility still lag behind the complexity of the design space. LEAF, OARF, and libraries such as Opacus help, but the literature remains fragmented across metrics, privacy accountants, sampling assumptions, attack protocols, and heterogeneous participation patterns [182]-[184]. As a result, method comparisons are often less decisive than individual papers suggest. ¹²

Challenges and Prospects

A first unresolved challenge is **semantic alignment between privacy guarantees and deployment goals**. Many papers report (ϵ, δ) -DP numbers without making clear whether the guarantee is record-

level, example-level, client-level, or user-level; whether composition assumes Poisson subsampling or fixed cohorts; and whether the server, the shuffler, or only the network links are trusted [15]-[30], [31]-[45], [80]-[102]. This ambiguity is not merely expository. Different notions imply different attack surfaces, different legal interpretations, and different utility costs. A central prospect for the field is thus better standards for reporting neighboring relations, accounting assumptions, participation models, and adversary observations alongside accuracy.

A second challenge is **privacy-utility-robustness tri-optimization**. Noise that supports DP can attenuate certain poisoning or backdoor signals, but it can also destabilize aggregation, weaken anomaly detectors, and mask malicious behavior. Conversely, robust aggregation, verification, or filtering can change the observable surface and affect privacy accounting [101]-[120], [132]-[140]. Recent systems that combine DP with secure aggregation, verifiability, or Byzantine tolerance are encouraging, but the theory for simultaneously optimizing all three dimensions remains incomplete [113]-[117]. Strong prospects lie in joint designs where privacy noise, compression, and robust aggregation are co-optimized rather than layered independently.

A third challenge concerns **foundation models, multimodality, and structure**. Large language models, speech models, graph neural networks, and multimodal prompt systems are precisely the workloads where FL is attractive and naive DP is weakest. Their parameter spaces are large, sparse, anisotropic, and highly sensitive to clipping [72], [73], [160]-[175]. Parameter-efficient tuning, adapter privatization, prompt learning, and hybrid public/private pretraining are promising directions, but the theory is still catching up with practice. Important open questions include how to define meaningful user-level privacy for retrieval-augmented and agentic pipelines, how to privatize personalization without nullifying adaptation, and how to audit leakage from distilled or prompt-based artifacts after training [167]-[175].

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A fourth challenge is **heterogeneous privacy allocation and fairness**. Clients and silos frequently differ in data volume, sensitivity, compute, dropout risk, and regulatory exposure. The literature now includes heterogeneous clipping, personalized budgets, dynamic privacy allocation, and across-silo user-level private learning [64]-[71], [87], [92]. Yet the normative question remains open: what constitutes a fair private-FL policy when clients have different sensitivities, resources, or downstream benefits? Future work will likely combine personalized optimization with budget markets, incentive design, or constrained multi-objective formulations that jointly optimize utility, participation, and risk [155], [156].

A fifth challenge is **evaluation realism and reproducibility**. Many private FL papers still rely on small benchmark models, unrealistically stable participation, or idealized threat models. Production evidence from Gboard is therefore especially valuable because it exposes the importance of public pretraining, device eligibility policies, monitoring, and practical accountant choices [44], [175]. More broadly, the field needs standardized attack suites, auditable privacy reports, deployment-oriented datasets, and systems benchmarks that measure not only top-line accuracy and ϵ , but also latency, dropouts, compression, memory, verification cost, and cross-round compositional drift [182]-[184], [192], [193]. 12

Taken together, the prospects are strong. The direction of travel is toward **private federated systems by design**: adaptive and architecture-aware clipping, trust-minimized aggregation, exact-distribution compression, structured parameter-efficient adaptation, and post-training privacy auditing. If these threads converge, the next generation of PPML systems may offer privacy guarantees that are not only formally correct, but also operationally meaningful and economically deployable.

Conclusion

Privacy-preserving machine learning with federated learning and differential privacy has evolved from a narrow modification of distributed SGD into a broad research program spanning statistical privacy, cryptographic trust reduction, distributed optimization, and application-aware systems design. The most important lesson from the literature is that federated learning is not inherently private in the formal sense; privacy arises only when the training and communication pipeline is explicitly designed around a clear adversary model and a precise notion of neighboring data [1]-[30], [101]-[140].

The most successful methods therefore combine several ideas: careful privacy granularity, clipping and accounting tailored to participation, secure aggregation or related trust-reduction mechanisms, robustness-aware aggregation, and model adaptation strategies such as distillation, personalization, or low-rank tuning. The literature now includes convincing demonstrations in healthcare, recommendation, graph learning, mobile language modeling, and emerging foundation-model workloads [141]-[175], [192], [193]. The remaining work is substantial, but the field is no longer asking whether private FL is possible. It is now asking how to make it tight, verifiable, personalized, multimodal, and deployable at scale. ¹⁴

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